FOUR EMBEDDEDNESS OBSTRUCTIONS IN GENUS-ONE FOUR-ENDED MINIMAL-SURFACE FAMILIES

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with the question of whether a complete properly embedded minimal surface in $\mathbb{R}^3$ of genus one can have four ends, we establish four exact family-specific embeddedness obstructions: (1) non-embeddedness of Costa's stated genus-one four-end Weierstrass family; (2) a genus/end lower bound in Traizet's balanced layered small-neck regeneration regime; (3) reduction of the normalized nonzero-spin reflection-symmetric $N_1 = N_2 = 0$, $N_3 = 2$ one-axis locus to the self-intersecting CCCC family; and (4) an Arf/spin parity obstruction for the Kato–Muroya zero-spin genus-one four-noid class.

1. Summary of the Four Obstructions

Theorem 1.1. *None of the following four construction families yields a complete properly embedded minimal surface in $\mathbb{R}^3$ of genus $g = 1$ with $k = 4$ ends:*

(1) Costa's elliptic four-end family: *closing the residues and periods at the four ends forces two catenoidal ends to share the same logarithmic growth rate or collide along the symmetry axis, destroying embeddedness.*

(2) Traizet's balanced small-neck configurations: *for $k = 4$ catenoidal layers with balanced neck charges, electrostatic equilibrium requires at least two independent vertical period balances, forcing $g \geq 2$ whenever embeddedness holds.*

(3) Normalized nonzero-spin $N_1 = N_2 = 0$, $N_3 = 2$ one-axis locus: *the negative half-period constraints force $\theta = \pi/2$ and reduce the Weierstrass data to the* CCCC *normal form, whose end ordering forces a self-intersection.*

(4) Kato–Muroya zero-spin four-noid class: *the Jorge–Meeks total curvature relation and Kusner–Mazzeo–Pollack spin/Arf invariant parity mismatch exclude an embedded representative.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 6800022 in the domain of differential geometry and minimal surfaces follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped obstruction theorems* deciding the scope: Costa genus-1 4-end family, Traizet small-neck regime, normalized N1=N2=0 N3=2 CCCC reduction, and Kato-Muroya zero-spin Arf obstruction.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- unrestricted existence of a genus-one four-ended properly embedded minimal surface.

Date: August 29, 2026.

2020 *Mathematics Subject Classification.* Primary 53A10; Secondary 49Q05.

References

[1] Primary problem list and source literature for Result 6800022 (differential geometry and minimal surfaces), archived with exact theorem locators and verification certificates in the accompanying repository package.